HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Mathematics

F18CD

Multivariable Calculus and Analysis A

IN-PERSON EXAMINATION

Specimen Paper

Duration: 2 hours

Answer All Questions

Attached: No additional attachments
Supplied: No additional material supplied
Calculator Permitted: Yes

Ethical Commitment: By submitting answers to this paper, I declare that these answers are entirely my own work and have not been shared, in part or in whole or in any form, with any other student, or disseminated in any way.

I also confirm that I have reviewed and understand all University regulations, policies and procedures on assessment misconduct, including but not limited to accessing unauthorised materials during this examination or the use of unauthorised electronic devices.

I understand that infringing this statement would represent an academic misconduct offence subject to disciplinary action according to the University Regulations and Policy regarding Academic Misconduct, with possible significant consequences for degree progression and final degree outcome.

DO NOT REMOVE THIS PAPER FROM THE EXAM HALL

1. (a) Write down the definition of the supremum of a non-empty set $A \subset \mathbb{F}$, where $\mathbb{F}$ is an ordered field.

[3 marks]

(b) Let A be a non-empty subset of $\mathbb{R}$ that is bounded above. Prove that the supremum of A is unique.

[7 marks]

(c) Let A be the set

$$A = \left\{ \frac{n}{n+2} : n \in \mathbb{N} \right\}.$$

Prove that the supremum of A is 1 and the infimum of A is $1/3$. Is 1 the maximum of A and $1/3$ the minimum of A?

[7 marks]

(d) Let $\mathbb{I} := \mathbb{R} \setminus \mathbb{Q}$ denote the set of irrational numbers. Does $\mathbb{I}$ form a complete ordered field? If so, provide a proof. Otherwise, provide a counterexample.

[3 marks]

2. (a) Write down the definition of a convergent sequence (a_n) to converge to $a \in \mathbb{R}$.

[3 marks]

(b) Let (a_n) and (b_n) be convergent sequences with $\lim(a_n) = a$ and $\lim(b_n) = b$. Prove that

$$\lim(a_n b_n) = ab.$$

[7 marks]

(c) Consider the sequences (a_n) and (b_n) given by

$$a_n = \frac{5n}{4n+4}$$

and

$$b_n = \frac{2n^2 - n}{6n^2 - 3}.$$

Using the algebra of limits, provide guesses for what (a_n) and (b_n) converge to and then explicitly prove that both sequences indeed converge to your guesses directly from the definition of a convergent sequences.

[7 marks]

(d) Let (a_n) be a convergent sequence. Does it follow that (a_n) is monotone? If so, provide a proof. Otherwise, provide a counterexample.

[3 marks]

3. (a) Write down the definition of a conditionally convergent series.

[3 marks]

(b) Prove that a series $\sum_{k=1}^{\infty} a_k$ is convergent if and only if for every $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that

$$|a_{m+1} + a_{m+2} + \cdots + a_n| < \varepsilon$$

for every $n > m > N$.

[7 marks]

(c) Consider the series

$$\sum_{k=1}^{\infty} \frac{2\sin(k) + \cos(k)}{k^7}.$$

Prove that this series is convergent.

[4 marks]

(d) Prove that the series

$$\sum_{k=0}^{\infty} \frac{(-1)^k \sqrt{k^2+4}}{k^2 + 5k + 4}$$

is conditionally convergent.

[6 marks]

4. (a) Write down the definition of a limit point $c \in \mathbb{R}$ of a set $D \subset \mathbb{R}$.

[3 marks]

(b) Let $f : D \to \mathbb{R}$ and $c \in \overline{D}$. Prove that $\lim_{x \to c}(f(x)) = L$ if and only if every sequence (a_n) with $\lim(a_n) = c$ and $a_n \in D \setminus \{c\}$ satisfies $\lim(f(a_n)) = L$.

[7 marks]

(c) Prove that the function $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^2 + x - 2$ is continuous at $c = 2$.

[6 marks]

(d) Let $p : \mathbb{R} \to \mathbb{R}$ given by

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0$$

where n is an odd number greater than zero. Does it follow that $p(x)$ always has at least one real root? If so, provide a proof. Otherwise, provide a counterexample.

[4 marks]

5. (a) Write down the definition of a function $f : D \to \mathbb{R}$ to be differentiable at a point $c \in D$.

[3 marks]

(b) Let $\sum_{k=0}^{\infty} a_k x^k$ be a power series with $a_k \neq 0$ for all $k \in \mathbb{N} \cup \{0\}$. Prove that if

$$\lim \left| \frac{a_{k+1}}{a_k} \right| = L,$$

with $L \in (0, \infty)$, the radius of convergence is $1/L$. Otherwise, prove that if $L = 0$, the radius of convergence is ∞.

[7 marks]

(c) Find the radius of convergence of the power series

$$\sum_{k=0}^{\infty} \frac{(3k)!}{(k!)^3} x^k.$$

[6 marks]

(d) Let $f : D \to \mathbb{R}$ such that f is differentiable at the point $c \in D$. Does it follow that the function is second differentiable at the point $c \in D$? If so, provide a proof. Otherwise, provide a counterexample.

[4 marks]